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ME573 – Heat Transfer

Fall 2023

**Material Properties of Heat Sinks and the Effect on Final Temperature
using Finite Element Analysis**

Introduction

Heat sinks are a form of temperature control for microelectronics and circuits. Their purpose is to increase the surface area and therefore the flow of heat-transferring liquid, typically air, across the fins. This is a very important function to prevent circuits from frying or overheating due to the absence of heat dissipation. Heat sinks are considered heat exchangers and are often no larger than tens of millimeters in diameter.

Over the years, many different orientations, shapes and sizes of fins have been tested and hypothesized. The easiest to model and calculate for is a simple straight fin design with equal spacing between. When not factoring in geometry, materials of the heat sink is also a large factor in its ability to keep components cooled. An ideal heat sink would have a large thermal conductivity in order to conduct more heat to the environment. This experiment will utilize SolidWorks Thermal Analysis software to explore different materials for chips and heat sinks and how they affect the temperatures at different states of the heat transfer.

Experiment

Objective

The objective of this experiment is to identify T_1 , T_2 , and T_3 for each configuration using FEA software, as shown in Figure 1. The smallest T_3 values correlates to the best heat sink and chip combination.

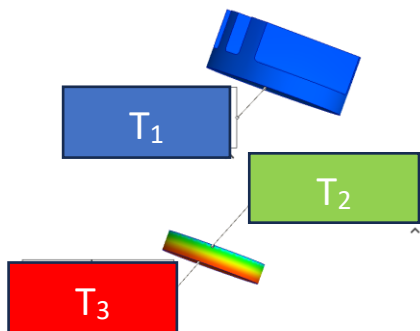


Figure 1 – Temperature labels of model

Design

The model being tested is a heat sink and microchip bonded together. The heat sink undergoes convection from the environment and contains three rectangular fins. The environment is assumed to be air at 300 Kelvin. The cylindrical microchip generates 25 W of power.

This experiment will test 9 different variations of the microchip-heatsink assembly. Each combination consists of a different chip material and heat sink material.

	Material	k (W/mK)
Heat Sink	1080 Alloy	200
	Alumina	30
	Copper	389
Chip	Porcelain Ceramic	1.4949
	Aluminum Nitride	250
	Gallium	275

Table 1

The combinations of the materials and the corresponding experiment number are listed in Table 2

Experiment #	Heat Sink Material	Chip Material
AX3	1060 Alloy	Porcelain Ceramic
AY3	1060 Alloy	Aluminum nitride
AZ3	1060 Alloy	Beryllia
BX3	Alumina	Porcelain Ceramic
BY3	Alumina	Aluminum nitride
BZ3	Alumina	Beryllia
CX3	Copper	Porcelain Ceramic
CY3	Copper	Aluminum nitride
CZ3	Copper	Beryllia

Table 2

Solid Model

The heat sink is modelled as a 70mm diameter cylinder with three fins extruded from the bottom surface. Each fin is 8 mm thick. The microchip is a simple 50mm diameter cylinder that is 10mm thick. The two components are mated concentrically and bonded on their flat faces. Both assemblies are displayed below.

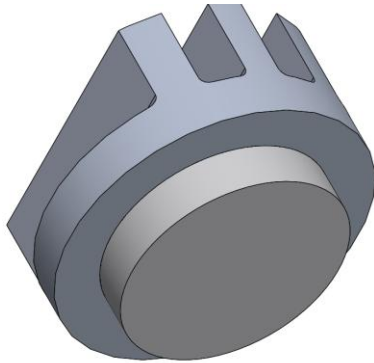


Figure 2

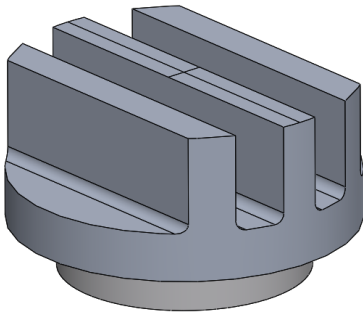


Figure 3

Boundary Conditions and Loading

The ambient environment is assumed to be air at 300 Kelvin with a convection coefficient of $h = 250 \text{ W/m}^2\text{K}$. Convection occurs on all exposed faces of the heat sink (Figure 4).

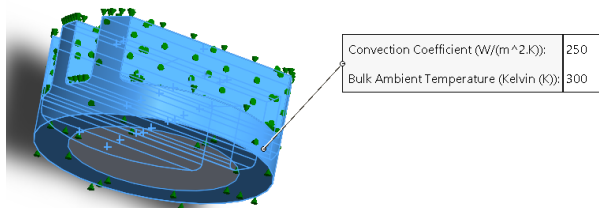


Figure 4

The two faces of the chip and the heat sink are bonded with a thermal resistance of $R_t = 0.1 \text{ K/W}$ (Figure 5).

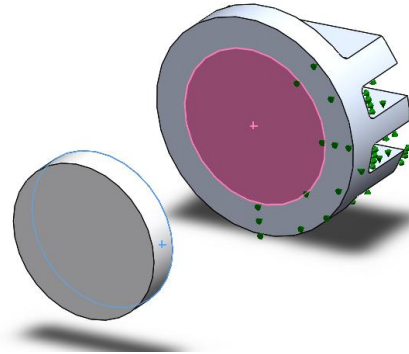


Figure 5

Heat power is applied to the entire chip with a value of 25 W (Figure 6).

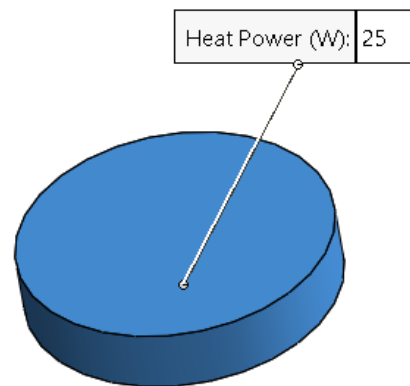


Figure 6

Mesh

Mesh control of 0.075 mm is applied to the fillets of the heat sink, as well as the entirety of the microchip since these are either areas of heat stress concentration or areas of interest (Figure 7 and Figure 8).

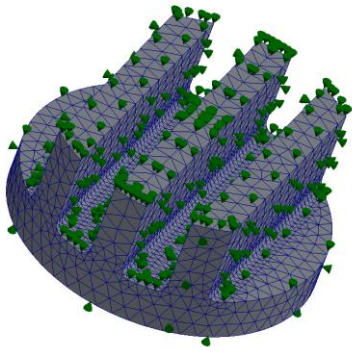


Figure 7

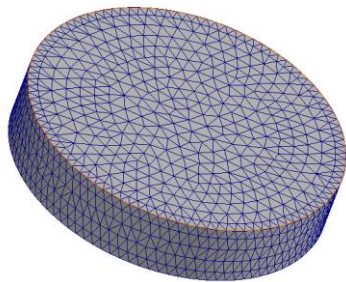


Figure 8

Results

The temperature plot of the assembly for all configurations are shown below. The heat maps are all set with a minimum value of 300K and a maximum value of 358K for comparison purposes (Figure 9, Figure 10, Figure 11, Figure 12, Figure 13, Figure 14, Figure 15, Figure 16, Figure 17).

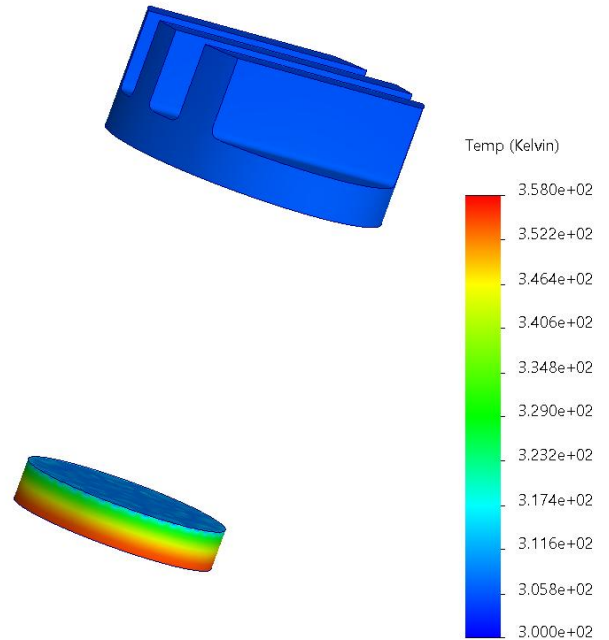


Figure 9 – Model AX3

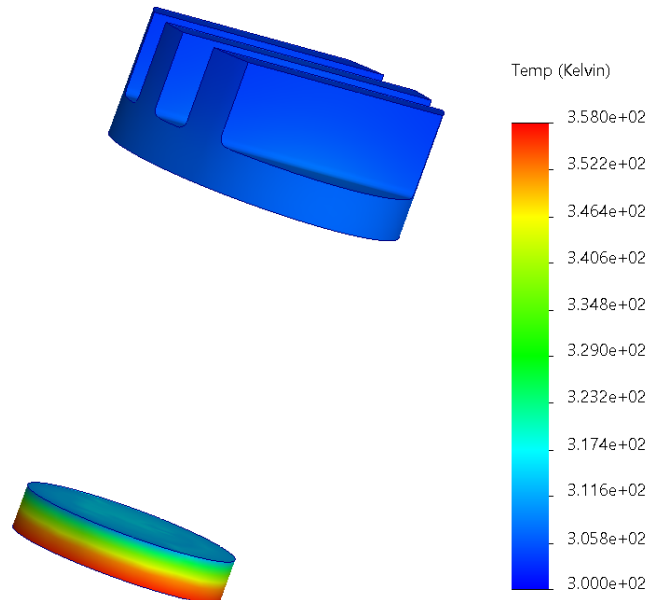


Figure 10 – Model AY3

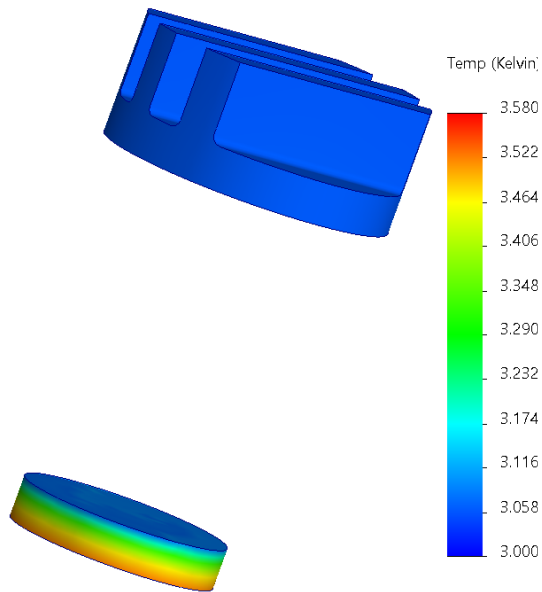


Figure 11 – Model AZ3

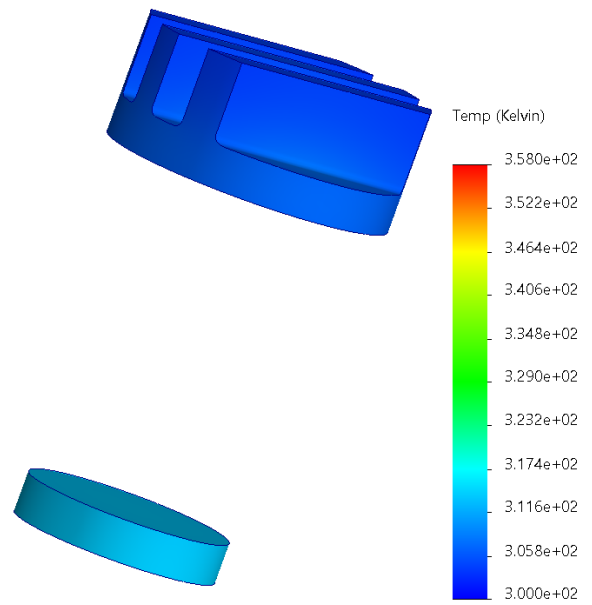


Figure 13 – Model BY3

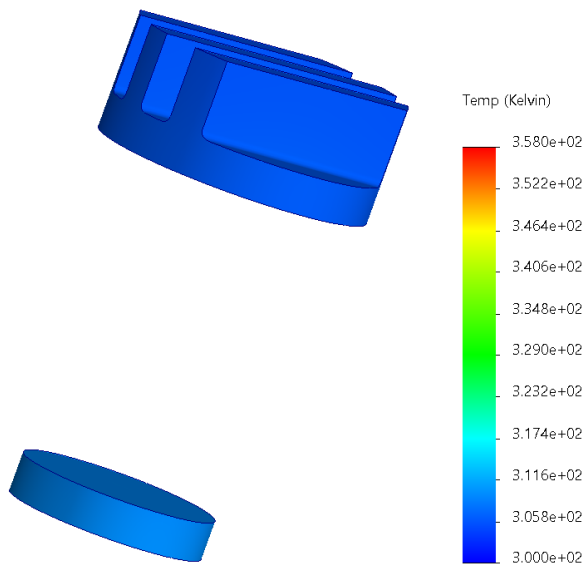


Figure 12 – Model BX3

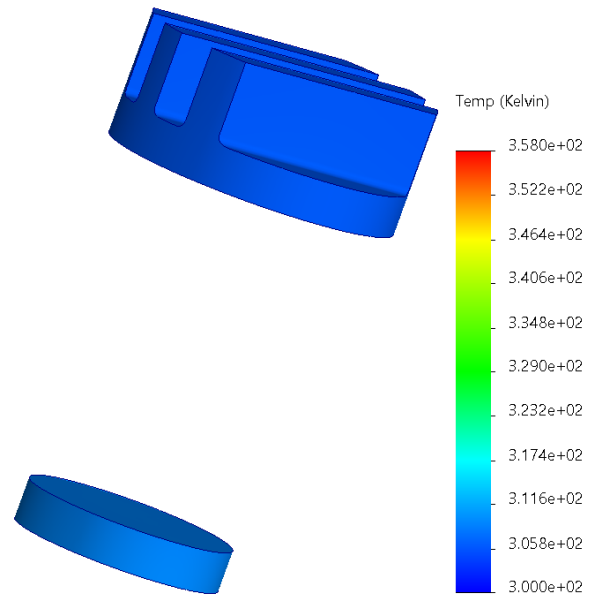


Figure 14 – Model BZ3

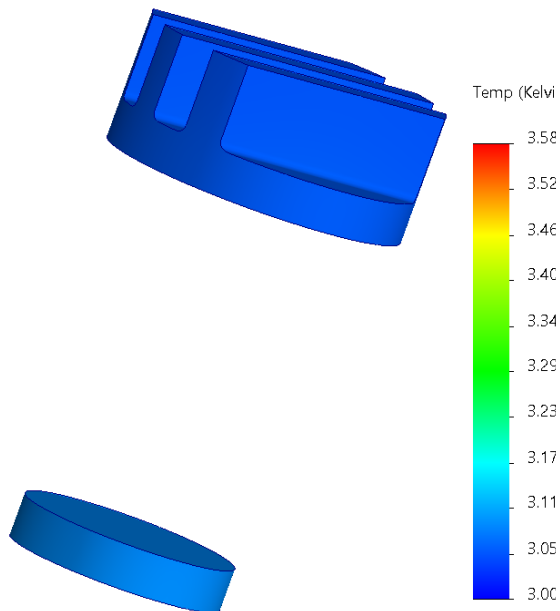


Figure 15 – Model CX3

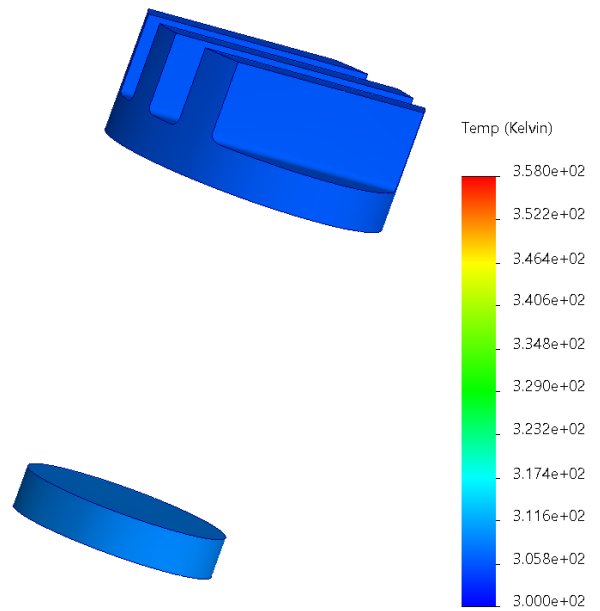


Figure 17 – Model CZ3

A summary of all SolidWorks results are listed in Table 3.

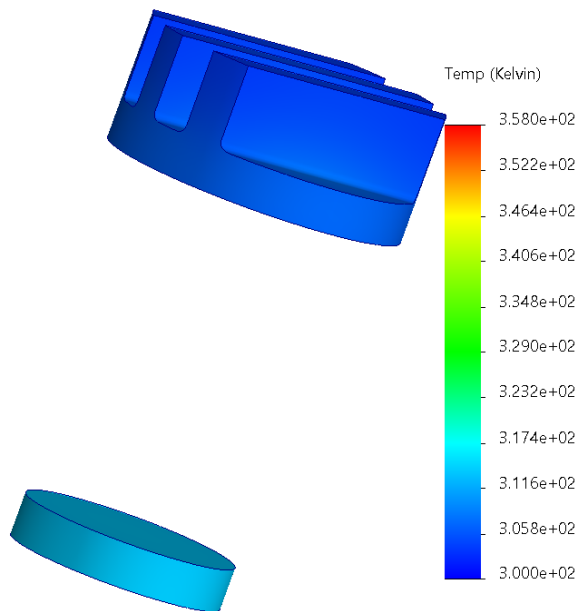


Figure 16 – Model CY3

Experiment #	T1 (K)	T2 (K)	T3 (K)
AX3	307.3	308.5	353.6
AY3	313.2	315.8	357.7
AZ3	306.7	309.4	352
BX3	307.3	309.6	309.8
BY3	312.2	313.9	314.1
BZ3	306.7	309.1	309.4
CX3	307.3	309.5	309.8
CY3	312.2	314.1	313.9
CZ3	306.7	309.1	309.3

Table 3 – Summary of FEA results

Analysis

Calculations

The temperature of the bottom surface of the chip (T_3) is calculated with Fourier's Law of Conduction and compared to the values obtained from the solid model. The following nodal analysis of the system is used:

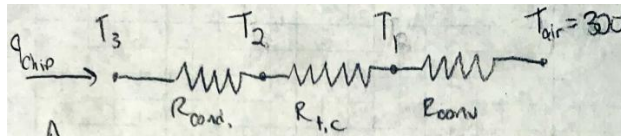


Figure 18 – Nodal analysis of model

The finned tabs of the heat sink must be considered in the convection calculation. The exposed surface area can be taken from the SolidWorks model and is found to be 0.01622 m². "q" is assumed to be 25W for the control volume around node 1, so all variables are known in order to solve for T_1 . The full calculation is attached in Appendix A.

Straight Fins

Rectangular^a

$$A_f = 2wL_c$$

$$L_c = L + (t/2)$$

$$A_p = tL$$

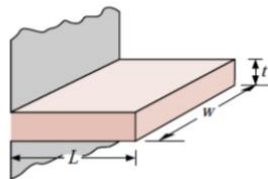


Figure 19 – Extended surface heat transfer

$$R_{t,conv} \equiv \frac{T_s - T_\infty}{q} = \frac{1}{hA}$$

Equation 1 – Convection

Conduction

$$q = -kA \frac{dT}{dx}$$

Equation 2 – Plane wall conduction and convection

Convection

$$q = hA(T_s - T_\infty)$$

Node 2 consists of contact resistance between the heat sink and the chip, which is taken to be 0.1 K/W.

Node 3 consists of conduction through the chip. The thermal conductivity of the material, as well as the cross-sectional area of 0.00196 m² is used

to find T_3 . The full calculation is attached in Appendix A.

$$R_{t,cond} \equiv \frac{T_{s,1} - T_{s,2}}{q_x} = \frac{L}{kA}$$

Equation 3 - Conduction

Data Validation

FEM vs. Calculated

The experimental and calculated temperature values are compared for each model, as well as the percent difference. Any error less than 2% is considered ideal and is highlighted in green. Any error above 10% is considered undesirable and is highlighted in red.

T1			
#	Calculated	FEM	% Difference
AX3	306.1652	307.3	-0.37064
AY3	306.1652	313.2	-2.2977
AZ3	306.1652	306.7	-0.17467
BX3	306.1652	307.3	-0.37064
BY3	306.1652	312.2	-1.97108
BZ3	306.1652	306.7	-0.17467
CX3	306.1652	307.3	-0.37064
CY3	306.1652	312.2	-1.97108
CZ3	306.1652	306.7	-0.17467

Table 4 – Percent difference for T_1

T2			
#	Calculated	FEM	% Difference
AX3	308.6652	308.5	0.05353
AY3	308.6652	315.8	-2.31149
AZ3	308.6652	309.4	-0.23805
BX3	308.6652	309.6	-0.30284
BY3	308.6652	313.9	-1.69594
BZ3	308.6652	309.1	-0.14086
CX3	308.6652	309.5	-0.27045
CY3	308.6652	314.1	-1.76073
CZ3	308.6652	309.1	-0.14086

Table 5 - Percent difference for T_2

T3			
	Calculated	FEM	% Difference
AX3	351.3244	353.6	-0.64771
AY3	308.9203	357.7	-15.7904
AZ3	308.8971	352	-13.9538
BX3	351.3244	309.8	11.8194
BY3	308.9203	314.1	-1.6767
BZ3	308.8971	309.4	-0.16279
CX3	351.3244	309.8	11.8194
CY3	308.9203	313.9	-1.61196
CZ3	308.8971	309.3	-0.13042

Table 6 - Percent difference for T_3

Uncertainty Analysis

Some uncertainties in this simulation may stem from the mesh sizing. It is impossible to create an infinitely small mesh size, and finer mesh size results in slower computing time.

Additionally, finite element methods will always perform under ideal circumstances. While this is favorable in terms of reproducibility, it can sometimes deviate from real life results. There is always a risk in using FEA software for having unrealistic results.

In the calculation for T_1 , which is the temperature at the inner surface of the heat sink, I wonder if there is an effect from not considering the conduction within the heat sink. The geometry is not a perfect plane wall, and this inconsistency could lead to deviations in the hand calculation. Since each temperature is referenced to one another in the calculations, this would affect each value.

Conclusion

In conclusion, the best material combination to minimize the temperature of the microchip is a alumina heat sink and beryllia chip. The chip only heated up to 309.4K, and the hand-calculated math is within 1% of the experimental values for all three temperatures.

The majority of the temperatures were very comparable between the FEM model and the

calculated. This proves that SolidWorks Thermal Analysis is a viable simulation software to predict the temperature gradient of a model.

This experiment also provides a better understanding of conduction and convection in an assembly, the effects of material properties, and a real-world application of heat transfer.

References

"Heat Sink." Wikipedia, Wikimedia Foundation, 27 Nov. 2023, en.wikipedia.org/wiki/Heat_sink#Material.

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Appendix A

